Exercise series 3: Relational algebra

Introduction to databases

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Tool: Online relational algebra calculator (RelaX)

- https://dbis-uibk.github.io/relax/
- Likes spaces between everything
  - Not: σ *hd>250* and *price<2000*(PC)
  - But: σ *hd > 250* and *price < 2000* (PC)

Relation algebra Operations

- Single table:
  - **projection** (π)
  - **selection** (σ)
  - **rename** (ρ)
- Multiple tables, Same Attributes:
  - **union** (∪)
  - **intersect** (∩)
  - **substract** (-)
- Multiple tables, Different Attributes
  - **cartesian product or cross join** (×)
  - **join** (⋈)
  - **division** (÷)

+ COMBO CREW

Exercise 1

RelaX > Load dataset > **Database Systems The Complete Book - Exercise 2.4.1**

Product.maker	Product.model	Product.type
'A'	1001	'pc'
'A'	1002	'pc'
'A'	1003	'pc'
'A'	2004	'laptop'
'A'	2005	'laptop'
'A'	2006	'laptop'
'B'	1004	'pc'
'B'	1005	'pc'
'B'	1006	'pc'
'B'	2007	'laptop'
'C'	1007	'pc'
'D'	1008	'pc'
'D'	1009	'pc'
'D'	1010	'pc'
'D'	3004	'printer'
'D'	3005	'printer'
'E'	1011	'pc'
'E'	1012	'pc'
'E'	1013	'pc'
'E'	2001	'laptop'
'E'	2002	'laptop'
'E'	2003	'laptop'
'E'	3001	'printer'
'E'	3002	'printer'
'E'	3003	'printer'
'F'	2008	'laptop'
'F'	2009	'laptop'
'G'	2010	'laptop'
'H'	3006	'printer'
'H'	3007	'printer'

Printer.model	Printer.color	Printer.type	Printer.price
3001	TRUE	'ink-jet'	99
3002	FALSE	'laser'	239
3003	TRUE	'laser'	899
3004	TRUE	'ink-jet'	120
3005	FALSE	'laser'	120
3006	TRUE	'ink-jet'	100
3007	TRUE	'laser'	200

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

Laptop.model	Laptop.speed	Laptop.ram	Laptop.hd	Laptop.screen	Laptop.price
2001	2	2048	240	20.1	3673
2002	1.73	1024	80	17	949
2003	1.8	512	60	15.4	549
2004	2	512	60	13.3	1150
2005	2.16	1024	120	17	2500
2006	2	2048	80	15.4	1700
2007	1.83	1024	120	13.3	1429
2008	1.6	1024	100	15.4	900
2009	1.6	512	80	14.1	680
2010	2	2048	160	15.4	2300

Exercise 1

1) Find all PC model numbers together with the price
2) Find all PC information with RAM greater than 1gb
3) Find RAM and speed of all Laptops and PC’s
4) Find all information about PC with more than 200gb hard disk costing less than 1000
5) Find all PC models with speed different from 3.2
6) Find all the possible combinations of a PC model and a printer
7) Find all PC models and the name of the maker
8) Find all makers that make all types of products
9) Find all PC and Laptop models with the same hd dimension

1.1) Find all PC model numbers together with the price

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

1.1) Find all PC model numbers together with the price

Projection!

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

π model,price (PC)

PC.model	PC.price
1001	2114
1002	995
1003	478
1004	649
1005	630
1006	1049
1007	510
1008	770
1009	650
1010	770
1011	959
1012	649
1013	529

1.2) Find all PC information with RAM greater than 1gb

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

1.2) Find all PC information with RAM greater than 1gb

Selection!

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

$\sigma_{ram > 1000}(PC)$

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1004	2.8	1024	250	649
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649

NB: you can use " < " even for text attributes!

1.3) Find RAM and speed of all Laptops and PC’s

Laptop.model	Laptop.speed	Laptop.ram	Laptop.hd	Laptop.screen	Laptop.price
2001	2	2048	240	20.1	3673
2002	1.73	1024	80	17	949
2003	1.8	512	60	15.4	549
2004	2	512	60	13.3	1150
2005	2.16	1024	120	17	2500
2006	2	2048	80	15.4	1700
2007	1.83	1024	120	13.3	1429
2008	1.6	1024	100	15.4	900
2009	1.6	512	80	14.1	680
2010	2	2048	160	15.4	2300

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

1.3) Find RAM and speed of all Laptops and PC's

Projection

π speed, ram (Laptop)

Laptop.speed	Laptop.ram
2	2048
1.73	1024
1.8	512
2	512
2.16	1024
1.83	1024
1.6	1024
1.6	512

π speed, ram (PC)

PC.speed	PC.ram
2.66	1024
2.1	512
1.42	512
2.8	1024
3.2	512
3.2	1024
2.2	1024
2.2	2048
2	1024
2.8	2048
1.86	2048
3.06	512

1.3) Find RAM and speed of all Laptops and PC's

Projection + Union! $\pi_{speed,\ ram}(Laptop) \cup \pi_{speed,\ ram}(PC)$

Laptop.speed	Laptop.ram
2	2048
1.73	1024
1.8	512
2	512
2.16	1024
1.83	1024
1.6	1024
1.6	512

∪

PC.speed	PC.ram
2.66	1024
2.1	512
1.42	512
2.8	1024
3.2	512
3.2	1024
2.2	1024
2.2	2048
2	1024
2.8	2048
1.86	2048
3.06	512

→

Laptop.speed	Laptop.ram
2	2048
1.73	1024
1.8	512
2	512
2.16	1024
1.83	1024
1.6	1024
1.6	512
2.66	1024
2.1	512
1.42	512
2.8	1024
3.2	512
3.2	1024
2.2	1024
2.2	2048
2	1024
2.8	2048
1.86	2048
3.06	512

1.4) Find all information about PC with more than 200gb hard disk costing less than 1000

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

1.4) Find all information about PC with more than 200gb hard disk costing less than 1000

Selection

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770

σ hd > 200 (PC)   σ price < 1000 (PC)

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

1.4) Find all information about PC with more than 200gb hard disk costing less than 1000

Selection + Intersection!

σ hd > 200 (PC) ∩ σ price < 1000 (PC)

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	[illegible]	[illegible]
1008	2.2	2048	[illegible]	[illegible]
1009	2	1024	[illegible]	[illegible]
1010	2.8	2048	[illegible]	[illegible]

∩

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1007	[illegible]	1024	200	510
[illegible]	[illegible]	2048	250	770
[illegible]	[illegible]	1024	250	650
[illegible]	[illegible]	2048	300	770
[illegible]	[illegible]	2048	160	959
[illegible]	[illegible]	1024	160	649
[illegible]	[illegible]	512	80	529

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1002	2.1	512	250	995
1004	2.8	1024	250	649
1005	3.2	512	250	630
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770

1.4) Find all information about PC with more than 200gb hard disk costing less than 1000

Selection + Intersection!

$\sigma_{hd > 200}(PC) \cap \sigma_{price < 1000}(PC)$

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	[illegible]	[illegible]
1008	2.2	2048	[illegible]	[illegible]
1009	2	1024	[illegible]	[illegible]
1010	2.8	2048	[illegible]	[illegible]

∩

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
[illegible]	[illegible]	1024	200	510
[illegible]	[illegible]	2048	250	770
[illegible]	[illegible]	1024	250	650
[illegible]	[illegible]	2048	300	770
[illegible]	[illegible]	2048	160	959
[illegible]	[illegible]	1024	160	649
[illegible]	[illegible]	512	80	529

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1002	2.1	512	250	995
1004	2.8	1024	250	649
1005	3.2	512	250	630
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770

(Or just selection…) $\sigma_{hd > 200 \wedge price < 1000}(PC)$

1.5) Find all PC models with speed different from 3.2

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

1.5) Find all PC models with speed different from 3.2

Selection

σ speed = 3.2 (PC)

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1005	3.2	512	250	630
1006	3.2	1024	320	1049

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1.5) Find all PC models with speed different from 3.2

Selection + Subtraction

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649

1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

PC - σ speed = 3.2 (PC)

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

1.5) Find all PC models with speed different from 3.2

Selection + Subtraction + Projection

PC.model
1001
1002
1003
1004

1007
1008
1009
1010
1011
1012
1013

π model (PC - σ speed = 3.2 (PC))

PC.model
1001
1002
1003
1004
1007
1008
1009
1010
1011
1012
1013

(Or… π model (σ speed ≠ 3.2 (PC))

1.6) Find all the possible combinations of a PC model and a printer model

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

Printer.model	Printer.color	Printer.type	Printer.price
3001	TRUE	'ink-jet'	99
3002	FALSE	'laser'	239
3003	TRUE	'laser'	899
3004	TRUE	'ink-jet'	120
3005	FALSE	'laser'	120
3006	TRUE	'ink-jet'	100
3007	TRUE	'laser'	200

1.6) Find all the possible combinations of a PC model and a printer model

Projection

π model (PC)   π model (Printer)

PC.model
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012
1013

Printer.model
3001
3002
3003
3004
3005
3006
3007

1.6) Find all the possible combinations of a PC model and a printer model

Projection + Cartesian Product!

π model (PC) × π model (Printer)

PC.model
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012
1013

×

Printer.model
3001
3002
3003
3004
3005
3006
3007

PC.model	Printer.model
1001	3001
1001	3002
1001	3003
1001	3004
1001	3005
1001	3006
1001	3007
1002	3001
1002	3002
1002	3003

PC.model	Printer.model
	3004
	3005
	3006
	3007
	3001
	3002
	3003
	3004
	3005
	3006

PC.model	Printer.model
	3007
	3001
	3002
	3003
	3004
	3005
	3006
	3007
	3001
	3002

Etc…

1.7) Find all PC models and the name of the maker

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

Product.maker	Product.model	Product.type
'A'	1001	'pc'
'A'	1002	'pc'
'A'	1003	'pc'
'A'	2004	'laptop'
'A'	2005	'laptop'
'A'	2006	'laptop'
'B'	1004	'pc'
'B'	1005	'pc'
'B'	1006	'pc'
'B'	2007	'laptop'
'C'	1007	'pc'
'D'	1008	'pc'
etc…	etc….	etc…

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1.7) Find all PC models and the name of the maker

Join $PC \bowtie Product$

PC.model	PC.speed	PC.ram	PC.hd	PC.price	Product.maker	Product.type
1001	2.66	1024	250	2114	'A'	'pc'
1002	2.1	512	250	995	'A'	'pc'
1003	1.42	512	80	478	'A'	'pc'
1004	2.8	1024	250	649	'B'	'pc'
1005	3.2	512	250	630	'B'	'pc'
1006	3.2	1024	320	1049	'B'	'pc'
1007	2.2	1024	200	510	'C'	'pc'
1008	2.2	2048	250	770	'D'	'pc'
1009	2	1024	250	650	'D'	'pc'
1010	2.8	2048	300	770	'D'	'pc'
1011	1.86	2048	160	959	'E'	'pc'
1012	2.8	1024	160	649	'E'	'pc'
1013	3.06	512	80	529	'E'	'pc'

1.7) Find all PC models and the name of the maker

Join + Projection! π model, maker (PC $\bowtie$ Product)

PC.model	Product.maker
1001	'A'
1002	'A'
1003	'A'
1004	'B'
1005	'B'
1006	'B'
1007	'C'
1008	'D'
1009	'D'
1010	'D'
1011	'E'
1012	'E'
1013	'E'

1.7) Find all PC models and the name of the maker

Join + Projection!

π model, maker (PC $\bowtie$ Product)

PC.model	Product.maker
1001	'A'
1002	'A'
1003	'A'
1004	'B'
1005	'B'
1006	'B'
1007	'C'
1008	'D'
1009	'D'
1010	'D'
1011	'E'
1012	'E'
1013	'E'

Or …

π PC.model, Product.maker
(
σ PC.model = Product.model (PC × Product)
)

(no, thanks)

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1.8) Find all makers that make all types of products

Product.maker	Product.model	Product.type
etc…	etc…	etc…
'D'	3005	'printer'
'E'	1011	'pc'
'E'	1012	'pc'
'E'	1013	'pc'
'E'	2001	'laptop'
'E'	2002	'laptop'
'E'	2003	'laptop'
'E'	3001	'printer'
'E'	3002	'printer'
'E'	3003	'printer'
'F'	2008	'laptop'
etc…	etc…	etc…

1.8) Find all makers that make all types of products

Projection

π maker, type (Product)

Product.maker	Product.type
'A'	'pc'
'A'	'laptop'
'B'	'pc’
'B'	'laptop'
'C'	'pc'
'D'	'pc'
'D'	'printer'
'E'	'pc'
'E'	'laptop'
'E'	'printer'
'F'	'laptop'
'G'	'laptop'
'H'	'printer'

π type (Product)

Product.type
'pc'
'laptop'
'printer'

1.8) Find all makers that make all types of products

Projection + Division! π maker, type (Product) ÷ π type (Product)

Product.maker	Product.type
'A'	'pc'
'A'	'laptop'
'B'	'pc’
'B'	'laptop'
'C'	'pc'
'D'	'pc'
'D'	'printer'
'E'	'pc'
'E'	'laptop'
'E'	'printer'
'F'	'laptop'
'G'	'laptop'
'H'	'printer'

÷

Product.type
'pc'
'laptop'
'printer'

Product.maker
'E'

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1.9) Find all PC and Laptop models with the same hd dimension

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

Laptop.model	Laptop.speed	Laptop.ram	Laptop.hd	Laptop.screen	Laptop.price
2001	2	2048	240	20.1	3673
2002	1.73	1024	80	17	949
2003	1.8	512	60	15.4	549
2004	2	512	60	13.3	1150
2005	2.16	1024	120	17	2500
2006	2	2048	80	15.4	1700
2007	1.83	1024	120	13.3	1429
2008	1.6	1024	100	15.4	900
2009	1.6	512	80	14.1	680
2010	2	2048	160	15.4	2300

1.9) Find all PC and Laptop models with the same hd dimension

Projection

π model, hd (PC)

π model, hd (Laptop)

PC.model	PC.hd
1001	250
1002	250
1003	80
1004	250
1005	250
1006	320
1007	200
1008	250
1009	250
1010	300
1011	160
1012	160
1013	80

Laptop.model	Laptop.hd
2001	240
2002	80
2003	60
2004	60
2005	120
2006	80
2007	120
2008	100
2009	80
2010	160

1.9) Find all PC and Laptop models with the same hd dimension

Projection + Renaming

π model, hd (PC) ρ model2 $\leftarrow$ model (π model, hd (Laptop))

PC.model	PC.hd
1001	250
1002	250
1003	80
1004	250
1005	250
1006	320
1007	200
1008	250
1009	250
1010	300
1011	160
1012	160
1013	80

Laptop.model2	Laptop.hd
2001	240
2002	80
2003	60
2004	60
2005	120
2006	80
2007	120
2008	100
2009	80
2010	160

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1.9) Find all PC and Laptop models with the same hd dimension

Projection + Renaming + Join!

$$\pi \text{ model, hd (PC)} \bowtie \rho \text{ model2} \leftarrow \text{model } (\pi \text{ model, hd (Laptop)})$$

PC.model	PC.hd
1001	250
1002	250
1003	80
1004	250
1005	250
1006	320
1007	200
1008	250
1009	250
1010	300
1011	160
1012	160
1013	80

$\bowtie$

Laptop.model2	Laptop.hd
2001	240
2002	80
2003	60
2004	60
2005	120
2006	80
2007	120
2008	100
2009	80
2010	160

PC.model	PC.hd	Laptop.model2
1003	80	2002
1003	80	2006
1003	80	2009
1011	160	2010
1012	160	2010
1013	80	2002
1013	80	2006
1013	80	2009

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Assignment: A Useful Trick

- Sometimes it makes sense to store some tables to re-use them again and again, exactly what you would do with a variable in any programming language.
- This is not an operation belonging to pure relational algebra

EG: find and store all the computer models (PC or Laptop)

$$\text{Computer\_models} = (\pi_{\text{model}}\ \text{PC} \cup \pi_{\text{model}}\ \text{Laptop})$$

Assignment

Importing databases from text files

- Open the dialog for importing databases
- Click “create new dataset”
- Paste database contents
- Click “use Group in editor”
- Click on the “Relational Algebra” tab
- Done